PA 307: Logical Inference and Statistical Analysis

Department of Public Administration | Jahangirnagar University

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Class: Thursday 8:00–9:30 AM | Office Hours: Wednesday 10:00–11:00 AM

Course Description

Every policy argument is, at its core, an inference, a claim that certain premises, if true, support a particular conclusion. A minister argues that a subsidy will reduce poverty. A health official argues that a vaccine mandate will lower transmission. An NGO argues that a cash transfer program improved school attendance. These arguments look different on the surface, but they share a common structure: they reason from what is known toward what is claimed. The question this course teaches you to ask and answer is: Are you entitled to that conclusion?

In the first half, we study formal logic: the rules that govern what follows from what, independent of subject matter. You will translate natural-language policy arguments into symbolic form, test them with truth tables, identify when an argument is valid regardless of whether its premises are true, and name the informal fallacies that fill public discourse. In the second half, we study probability and statistical inference: the tools for reasoning carefully from data under uncertainty. You will describe data, model uncertainty with probability, and conduct and interpret hypothesis tests.

These halves are not separate courses bolted together. They are two expressions of the same underlying skill. Formal logic asks what must be true if our premises hold. Statistical inference asks what is probably true given our data. Both ask the same question: given what you have, what are you actually entitled to claim? Public policy is full of people claiming more than their evidence warrants. This course teaches you to recognize that and to avoid doing it yourself.

Learning Objectives

By the end of this course, students will be able to:

1. **Translate arguments:** convert natural-language policy arguments into symbolic logical form using standard connectives, accurately representing their logical structure.
2. **Evaluate validity and soundness:** use truth tables and proof strategies to determine whether an argument is valid, and distinguish validity from soundness.
3. **Identify reasoning errors:** recognize and name common informal fallacies in policy arguments and explain precisely why they fail.
4. **Describe data rigorously:** calculate and correctly interpret measures of central tendency and spread, and explain how summary statistic choices frame and sometimes distort policy arguments.
5. **Apply probability:** compute probabilities using the basic rules, interpret conditional probabilities and independence, and connect probability reasoning to the logic of conditionals.
6. **Conduct and interpret hypothesis tests:** formulate null and alternative hypotheses, apply one-sample, two-sample, and chi-square tests, and interpret results including p-values, confidence intervals, and error types.
7. **Communicate inferences professionally:** translate technical statistical findings into policy-facing language that is accurate, appropriately qualified, and decision-relevant.

Teaching Approach

Logic and statistics are cumulative technical skills. A student who does not understand truth tables will struggle with proof strategies. A student who does not understand conditional probability will struggle with p-values. This means there is no substitute for regular practice. The twelve weekly problem sets are the spine of this course, not supplementary exercises, but the primary mechanism by which you will develop competence. They are deliberately short enough to complete in one or two focused hours, but demanding enough that passive reading will not prepare you for them. You must work the problems to learn the material.

That said, practice without understanding is mechanical and fragile. Each session is designed around a conceptual anchor, the single idea that week's problems depend on, followed by an in-class exercise that forces you to apply it in a context you have not seen before. The in-class exercises are drawn almost entirely from public administration, policy analysis, and social science data, because the goal is inferential competence in the contexts where you will actually use these tools.

One thread runs through both halves of the course, and I name it explicitly from the first day: the relationship between evidence and entitlement. A valid logical argument entitles you to its conclusion if the premises are true. A well-designed hypothesis test entitles you to a qualified probabilistic inference if the assumptions hold. In both cases, the question is the same: given what you have, what are you actually entitled to claim? Learning to answer that question honestly is the entire point of this course.

Signature Assignments

Logic Analysis Paper (Individual) — 10% of Final Grade

The Logic Analysis Paper asks you to take a real policy argument from a publicly available source, a government document, a policy brief, a parliamentary speech, or a published editorial, and subject it to the formal logic tools developed in Units 1 and 2. The goal is to show that you can move from reading an argument to diagnosing its logical structure with precision.

The Task

Select a policy argument of at least moderate complexity, not a slogan, but a sustained piece of reasoning with identifiable premises and a conclusion. Your paper must:

- **Reconstruct the argument formally:** identify premises and conclusion, translate the argument into symbolic form, and draw the structure explicitly.
- **Evaluate validity:** use a truth table or proof to determine whether the conclusion follows necessarily from the premises. If invalid, identify the specific logical gap.
- **Evaluate soundness:** assess whether the premises are actually true, citing evidence where available. Show whether a valid argument is also sound.
- **Identify fallacies:** name any informal fallacies present, explain the flaw precisely, not just the label, and show how the argument could be repaired.
- **State the policy implication:** in one paragraph, explain what your logical analysis implies for how seriously this argument should be treated as a basis for policy action.

Format, Length, and Process

900–1,100 words. Use formal logical notation where appropriate, but every symbolic claim must be explained in plain language alongside it. The paper is assigned in Week 3 and due at the end of Week 6. In Week 4, bring your selected argument to class. We will spend fifteen minutes in pairs stress-testing each other's argument choices before you commit.

Evaluation Criteria

- Reconstruction accuracy: Is the argument's logical structure correctly identified and translated?
- Validity assessment: Is the truth table or proof correct, and the validity judgment accurate?
- Soundness assessment: Is the factual evaluation of premises evidence-based and honest about uncertainty?
- Fallacy identification: Are fallacies named precisely and explained specifically, not just labeled?
- Clarity: Is the analysis accessible to a non-logician reader without sacrificing precision?

Team Data Interpretation Memo — 15% of Final Grade (+ 10% for Oral Presentation)

The Team Data Interpretation Memo asks your team to do what policy analysts actually do with statistics: take a real dataset on a public problem, describe what it shows, test a hypothesis about it, and write findings up as a professional memo for a decision-maker who needs to act. The companion oral presentation asks you to defend those findings under questioning.

The Task

Your team selects one dataset from an approved list distributed in Week 8. Datasets cover policy-relevant topics including public health outcomes, education attainment, labor market conditions, government service delivery, or public expenditure. Your memo must do all of the following:

- **State a specific, falsifiable research question:** not "is education related to income?" but "do districts with higher per-pupil spending have significantly lower primary school dropout rates?" Vague questions will be returned without a grade.
- **Describe the data:** produce appropriate descriptive statistics and at least two visualizations. Explain what the summary statistics reveal and what they conceal or distort.
- **Assess distributional assumptions:** identify any distributional assumptions your chosen hypothesis test requires and assess whether the data support them.
- **Conduct a hypothesis test:** state null and alternative hypotheses, select and apply the appropriate test, report the test statistic and p-value, and state your conclusion with explicit qualification.
- **Translate findings for a policy audience:** convert your statistical results into a recommendation or qualified inference that a non-statistician ministry official can understand and act on. Be explicit about what your analysis establishes and what it cannot.

Format, Length, and Milestones

The written memo should be 1,200–1,500 words plus supporting tables and figures, written to a hypothetical agency or ministry director. In Week 7 (midterm week), teams submit their chosen dataset and a one-sentence research question for approval. In Week 10, teams submit a one-page outline: dataset description, research question, and planned methods for instructor feedback. The full written memo is due at the end of Week 14. Oral presentations take place in Sessions 14 and 15 (10–12 minutes per team, with open questions from the class and instructor).

Evaluation Criteria: Written Memo

- Research question: Is it specific, falsifiable, and policy-relevant?
- Descriptive analysis: Are statistics and visualizations accurate, appropriate, and interpretively honest?
- Hypothesis test: Is the correct test selected, correctly applied, and accurately interpreted, including appropriate qualification of the conclusion?
- Policy translation: Are findings communicated accurately and accessibly for a non-technical reader?
- Professional presentation: Is the memo clear, well-organized, and written at an appropriate level for a ministry audience?

Evaluation Criteria: Oral Presentation

- Clarity: Are findings explained accurately and without unnecessary jargon?
- Precision under questioning: Does the team respond to questions with intellectual honesty, including acknowledging the limits of the analysis?
- Qualifications: Does the team avoid overstating what the data show?

Weekly Problem Sets — 30% of Final Grade

Twelve problem sets are distributed across the semester, one for each week of new content (none during the midterm week). Each set contains 5–8 problems requiring application of that session's concept. The two lowest scores are dropped automatically. This policy exists to accommodate illness and emergencies, not to justify skipping sets.

Problem sets are individual work. You may discuss the approach to a problem with classmates, but every written solution must be your own, drafted independently. Write the names of any classmates you worked with at the top of your submission. On any problem that involves reasoning rather than pure computation, write at least one sentence explaining your reasoning. A correct numerical answer with no justification receives at most half credit in this course. Showing your work is not optional formality. It is evidence that you understand what you are doing.

Problem sets are due at the start of the following Thursday class. Solutions are posted after the due deadline. Late submissions after solutions are posted cannot be accepted for credit.

Assessment and Grading

Assessment	Weight	What Is Being Evaluated
Participation	5%	Quality of engagement in in-class exercises and problem-solving sessions. Showing work and reasoning aloud counts more than arriving at the right answer silently.
Weekly Problem Sets — 12 total, drop lowest 2	30%	Weekly sets of 5–8 problems applying that session's logic or statistics concept. Lowest 2 dropped. Individual work; collaboration on approach is permitted but written

		solutions must be your own. Due at the start of the following Thursday class.
Logic Analysis Paper (Individual, ~900 words)	10%	Written reconstruction and evaluation of a real policy argument using formal logic tools from Units 1–2. Evaluated on precision of reconstruction, validity assessment, and clarity of reasoning.
Team Data Interpretation Memo (Team, ~1,200 words)	15%	Team analysis of a real policy dataset: descriptive statistics, probability assessment, hypothesis test, and policy recommendation. Written as a professional memo to a hypothetical decision-maker. Evaluated on methods accuracy, interpretive honesty, and quality of communication.
Team Oral Presentation and Defense	10%	10-minute public presentation of the team memo findings with open questions. Evaluated on clarity, accuracy under questioning, and appropriateness of qualifications.
Midterm Exam	15%	Covers Units 1–2 (logic). Truth table problems, argument translation, validity assessment, and two short analytical response questions. One sheet of notes permitted.
Final Exam	15%	Cumulative, emphasis on Units 3–5 (statistics). Computation problems and interpretation questions applied to a novel dataset. Open note. Calculator required.

Grading follows the university's standard grading policy. Late submissions are penalized 10% per day unless an extension is requested at least 48 hours in advance. The two lowest problem set scores are dropped automatically. There is no extra credit.

Course Materials

This course draws from two primary texts, both freely available online. For the logic half, we use Chapter 1 of Daniel J. Velleman's How to Prove It: A Structured Approach (2nd ed., Cambridge University Press, 2006). For the statistics half, we use Chapters 1–11 of Barbara Illowsky and Susan Dean's Introductory Statistics (1st ed., OpenStax, 2013). Both texts are available as free PDFs, and links are posted on the course website.

Additional readings, such as policy documents, published empirical analyses, datasets and examples of flawed argumentation in public discourse, will be distributed on the course website before the relevant session. A calculator is required from Week 8 onwards. Statistical software is not required, but students using Excel or similar tools for the team project should note this in their memo.

Course Schedule

The course meets every Thursday, 8:00–9:30 AM, for 15 sessions over the semester.

Wk	Session	Topic, Reading, and In-Class Activities	Problem Set / Milestone
UNIT 1 — The Architecture of Arguments: Sentential Logic [Velleman Ch. 1.1–1.5]			
1	Thu	Course introduction. Central question: what makes an argument valid — and why does that matter for public policy? Distinguishing premises, conclusions, validity, and soundness. In-class exercise: students bring a real policy claim and informally assess whether the reasoning holds. Preview of the course arc: from formal logic to statistical inference as two expressions of the same skill — disciplined entitlement to a conclusion. [Velleman: Introduction]	**PS 1 distributed. Teams assigned (3–4 students).**
2	Thu	Logical connectives: conjunction, disjunction, negation, conditional, and biconditional. Truth conditions for each. Translating natural-language policy arguments into symbolic form. In-class exercise: five real policy statements translated into logical notation and checked for ambiguity. Why 'if the policy is adopted, outcomes will improve' is a much weaker claim than it sounds. [Velleman Ch. 1.1]	**PS 1 due. PS 2 distributed.**
3	Thu	Truth tables: constructing and reading them for compound statements. Logical equivalence, tautologies, and contradictions. Testing argument validity with truth tables. The conditional and biconditional in detail. In-class exercise: build truth tables for three policy-relevant conditional arguments and determine which are tautologies. [Velleman Ch. 1.2, 1.5]. LOGIC ANALYSIS PAPER ASSIGNED.	**PS 2 due. PS 3 distributed. Logic Analysis Paper assigned.**
UNIT 2 — Quantifiers, Proofs, and the Limits of Deduction [Velleman Ch. 1.3–1.4, 2.1–2.2, 3.1–3.2]			
4	Thu	Variables, sets, and quantifiers. Universal claims ('all bureaucrats satisfice') versus existential claims ('some bureaucrats satisfice'): why the distinction matters for policy generalization. Quantifier scope and how misreading it produces policy fallacies. In-class exercise: identify and correct quantifier errors in five real policy statements. [Velleman Ch. 1.3–1.4, 2.1–2.2]	**PS 3 due. PS 4 distributed.**
5	Thu	Proof strategies: direct proof, proof by contrapositive, proof by contradiction. What does it mean to prove something versus merely assert it? Why 'correlation does not imply causation' is a logical claim, not just a statistical warning. Students work through structured proofs of three policy-relevant propositions. In-class exercise: identify which proof strategy each argument implicitly uses and whether it succeeds. [Velleman Ch. 3.1–3.2]	**PS 4 due. PS 5 distributed.**
6	Thu	Inductive reasoning and its limits. Deductive validity versus inductive strength — and why the difference is critical for evidence-based policy. Common informal fallacies in policy argumentation: hasty generalization, false dichotomy, slippery slope, post hoc ergo propter hoc, appeal to authority. In-class argument critique: students receive three real policy arguments	**PS 5 due. LOGIC ANALYSIS PAPER DUE.**

		and must identify every logical flaw by name, explain it precisely, and propose a repair. LOGIC ANALYSIS PAPER DUE.	
7	Thu	MIDTERM EXAM — Units 1 and 2. The exam covers sentential logic, truth tables, quantifiers, proof strategies, and informal fallacies. Format: truth table construction (30%), argument translation and validity assessment (40%), two short analytical response questions requiring written justification (30%). A correct answer without reasoning receives at most half credit. One sheet of notes permitted. No problem set this week.	**MIDTERM EXAM. No PS this week. Team dataset selection due.**
UNIT 3 — From Logic to Data: Sampling, Description, and Probability [Illowsky & Dean Ch. 1–3]			
8	Thu	Bridge session: from logical inference to statistical inference. What is the relationship between a deductive argument and a statistical hypothesis? How does data constrain what we are entitled to claim? Key terms: population, sample, parameter, statistic. Types of data, sampling design, levels of measurement, and why sampling decisions are also logical decisions about what conclusions can follow. [Illowsky & Dean Ch. 1]. TEAM DATA MEMO PROJECT ASSIGNED.	**PS 6 distributed. Team Data Memo Project assigned.**
9	Thu	Descriptive statistics: stem-and-leaf plots, histograms, and box plots. Measures of central tendency: mean, median, mode — when each is appropriate and how each frames a policy argument differently. Measures of spread: range, variance, standard deviation, interquartile range. Skewness. In-class exercise: the same public expenditure dataset summarized three ways — mean versus median versus mode — and what story each tells to a minister seeking justification for a budget cut. [Illowsky & Dean Ch. 2]	**PS 6 due. PS 7 distributed.**
10	Thu	Probability: terminology and basic rules. Independent and mutually exclusive events. The two rules of probability: addition and multiplication. Conditional probability and contingency tables. Tree diagrams and Venn diagrams. Connecting probability to the logic of conditionals: $P(B\|A)$ is the probabilistic analogue of 'if A then B,' but they are not equivalent — a point with major implications for how we interpret evidence. In-class exercise: calculate conditional probabilities from a public health surveillance dataset and translate each result into a plain-language claim. [Illowsky & Dean Ch. 3]	**PS 7 due. PS 8 distributed. Team outline due: one page — dataset, research question, planned methods.**
UNIT 4 — The Language of Uncertainty: Distributions and the Central Limit Theorem [Illowsky & Dean Ch. 4, 6–7]			
11	Thu	Discrete random variables: probability distribution functions, expected value, standard deviation. The binomial distribution and its public policy applications — modeling binary outcomes: employed/unemployed, program compliant/non-compliant, voted/did not vote. In-class exercise: compute the expected number of beneficiaries reached under three different	**PS 8 due. PS 9 distributed. Instructor feedback on team outline returned.**

		program targeting designs, each with a different compliance probability. [Illowsky & Dean Ch. 4]	
12	Thu	Continuous random variables and the normal distribution. Standard normal distribution and z-scores. The Central Limit Theorem: why the sampling distribution of the mean approaches normality regardless of the population distribution, and why this is the theoretical foundation for nearly all applied inference. In-class exercise: use the CLT to explain to a non-statistician why a random sample of 400 households is sufficient to estimate a district's school enrollment rate — and what happens when the sample is not random. [Illowsky & Dean Ch. 5–7]	**PS 9 due. PS 10 distributed.**
UNIT 5 — Hypothesis Testing: Inferential Reasoning from Data [Illowsky & Dean Ch. 8–11]			
13	Thu	Confidence intervals and hypothesis testing I. Estimating population means and proportions. Null and alternative hypotheses. Type I and Type II errors. P-values and significance thresholds. Why a p-value is a logical statement — it quantifies the probability of observing data at least this extreme if the null hypothesis were true. Why a p-value does not tell you the probability that the null is true. In-class exercise: interpret three published hypothesis tests from public policy research and identify what each p-value does and does not establish. [Illowsky & Dean Ch. 8–9]	**PS 10 due. PS 11 distributed.**
14	Thu	Hypothesis testing II. Two-sample tests: comparing means and proportions across groups. When do we compare two groups, and what design assumptions must hold? Chi-square tests: goodness-of-fit and test of independence for categorical data. In-class exercise: run a chi-square test of independence on a real education survey dataset and write a two-paragraph interpretation for a ministry audience. TEAM PRESENTATIONS begin. [Illowsky & Dean Ch. 10–11]	**PS 11 due. PS 12 distributed. TEAM MEMO WRITTEN REPORT DUE. Presentations begin.**
15	Thu	Remaining team presentations. Course synthesis: tracing the inference chain from Week 1 to Week 15 — from truth tables to hypothesis tests. The common thread: at every step, the question is the same — given what you have, what are you entitled to claim? Final discussion: what can statistics prove, what can it only support, and why does that distinction matter for public policy? How to use these tools without misrepresenting what they show. FINAL EXAM distributed.	**PS 12 due. FINAL EXAM distributed (take-home, 72 hours, open note). Due Sunday midnight.**

Required Readings by Unit

Units 1–2: Formal Logic and Proof [Velleman, How to Prove It, 2nd ed., 2006]

- Ch. 1.1: Deductive reasoning and logical connectives.
- Ch. 1.2: Truth tables for compound statements.
- Ch. 1.3–1.4: Variables, sets, and operations on sets.

- Ch. 1.5: The conditional and biconditional connectives.
- Ch. 2.1–2.2: Quantifiers and equivalences involving quantifiers.
- Ch. 3.1–3.2: Proof strategies: direct proof, contrapositive, and contradiction.
- Supplementary: Selected examples of logical fallacies in real policy speeches and documents (distributed in class, Week 6).

Unit 3: Sampling, Descriptive Statistics, and Probability [Illowsky & Dean, Introductory Statistics, 1st ed., 2013]

- Ch. 1: Definitions of statistics and probability; data types; sampling design; levels of measurement.
- Ch. 2: Descriptive statistics: graphical displays, measures of central tendency and spread, skewness.
- Ch. 3: Probability: terminology, basic rules, independence, conditional probability, contingency tables.

Unit 4: Distributions and the Central Limit Theorem

- Ch. 4: Discrete random variables, probability distribution functions, expected value, binomial distribution.
- Ch. 5: Continuous random variables and continuous probability functions.
- Ch. 6: The normal distribution and the standard normal.
- Ch. 7: The Central Limit Theorem for sample means.

Unit 5: Confidence Intervals and Hypothesis Testing

- Ch. 8: Confidence intervals for single population means and proportions.
- Ch. 9: Hypothesis testing with one sample: null and alternative hypotheses, Type I and II errors, p-values.
- Ch. 10: Hypothesis testing with two samples — means and proportions.
- Ch. 11: The chi-square distribution: goodness-of-fit and test of independence.
- Supplementary: Gelman, A. & Loken, E. (2014). The Statistical Crisis in Science. American Scientist, 102(6): 460–465. On p-value misuse and what it means for policy evidence. (Distributed Week 13.)

Course Policies

Academic Integrity

All work submitted must be your own unless explicitly designated as team work. On problem sets, discussing approaches with classmates is permitted, but written solutions must be drafted independently. Identical or near-identical solutions submitted by different students are an integrity violation. Copying another student's work will be reported to the university and may result in a grade of zero or an F in the course. In a course whose entire purpose is teaching you to reason carefully and honestly, the integrity of your own arguments is not incidental. It is the subject matter.

Attendance

The course meets once a week. Each session builds directly on the previous one; missing a Thursday creates a gap that subsequent problem sets will expose. More than two unexcused absences will substantially reduce your participation grade. If you must miss a session, notify the instructor in advance and obtain notes from a classmate. You are responsible for all material covered whether present or absent.

Problem Set Policy

Problem sets are due at the start of Thursday class. Solutions are posted after the deadline. Late submissions before solutions are posted are penalized 10% per day. Submissions after solutions are posted cannot be accepted. The two lowest scores are dropped. This policy covers illness and emergencies. It is not an invitation to skip sets strategically, since problem sets are the primary practice mechanism of the course and missed practice will show in the exams.

Calculators and Software

A basic scientific calculator is required from Week 8 onwards and for the final exam. Statistical software (Excel, Stata, SPSS) is not required but may be used for the team project. The final exam is open note and will require computation.

Communication

Course materials, problem sets, and solutions are posted on the course website. Check regularly. For questions about course content or assignments, use office hours (Wednesday 10:00–11:00 AM) or email. I respond to emails within 48 hours on weekdays. Questions about problem set solutions should come during office hours, not via email. Working through a problem together is more useful than receiving an answer.